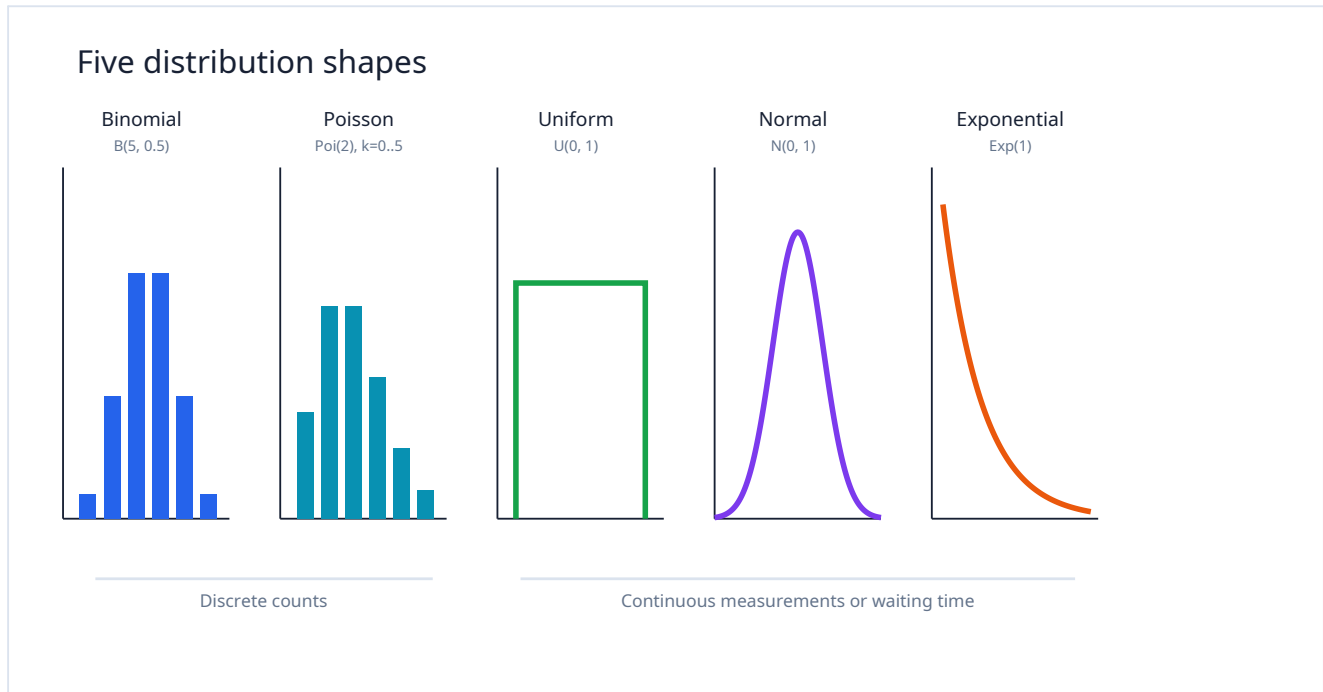
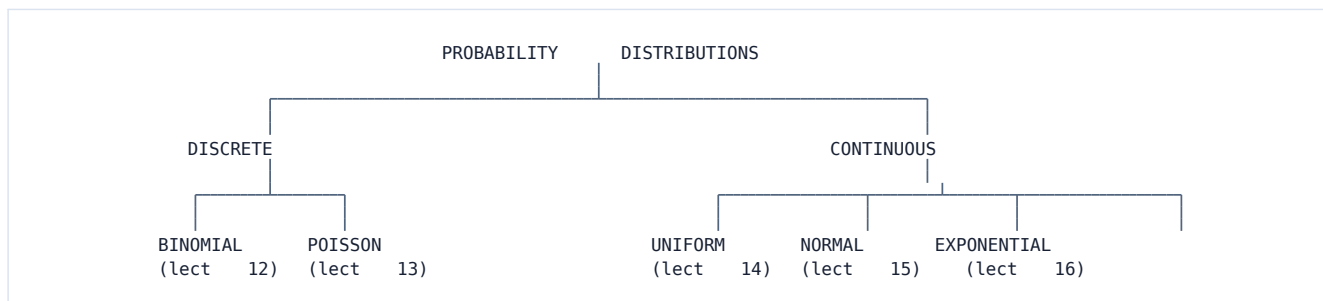


# 07 UNIFORM, NORMAL, EXPONENTIAL (ppt4, 44 pages, lectures 14-16)

## The family tree first (p001)



Five distribution shapes for comparison



## PART A: UNIFORM (p002-p006) - lecture 14

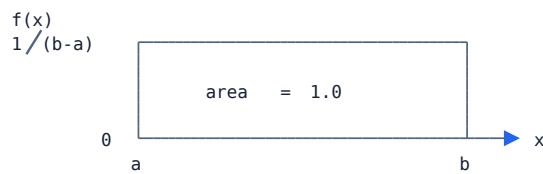
### 1. Definition

constant density, so equal-length subintervals have equal probabilities. Individual points each have probability zero.

$$f(x) = 1/(b-a) \quad a \leq x \leq b$$

$$f(x) = 0 \text{ otherwise}$$

the flat rectangle:



the area is width  $\times$  height  $= (b-a) \times 1/(b-a) = 1 \checkmark$

## 2. Properties

$$\mu = (a + b)/2$$

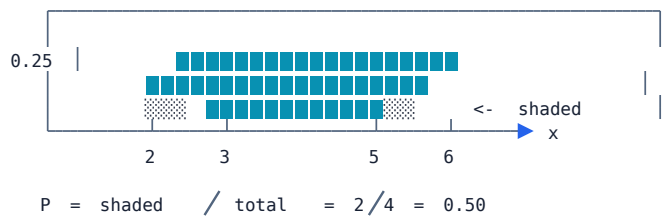
$$\sigma^2 = (b - a)^2/12 \text{ (the 12 is FIXED)}$$

THE LENGTH-RATIO SHORTCUT: probability = overlap length divided by total length. No integration needed.  
 $P(c \leq X \leq d) = \text{length}([c,d] \text{ intersection } [a,b]) / (b-a)$   
 If  $[c,d]$  lies inside  $[a,b]$ , this becomes  $(d-c)/(b-a)$ .

## 3. WORKED: U(2,6) (p004-p006)

|                                         |                   |
|-----------------------------------------|-------------------|
| $f(x) = 1/(6-2) = 0.25,$                | $2 \leq x \leq 6$ |
| $\mu$                                   | $= (2+6)/2 = 4$   |
| $\sigma^2 = (6-2)^2/12 = 16/12 = 1.333$ |                   |

$$P(3 < X < 5) = \text{length ratio} = (5-3)/(6-2) = 2/4 = 0.50$$

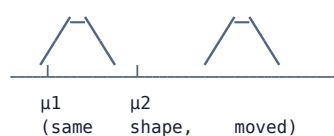


## PART B: NORMAL (p007-p037) - lecture 15

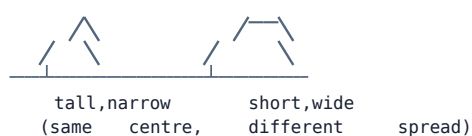
### 4. Shape facts (p008-p010)

BELL SHAPED  
 SYMMETRICAL  
 MEAN = MEDIAN = MODE =  $\mu$   
 range:  $-\infty$  to  $+\infty$   
 location set by  $\mu$ , spread set by  $\sigma$

changing  $\mu$  shifts:



changing  $\sigma$  widens:



## 5. The notation and the density (p010-p011)

$$X \sim N(\mu, \sigma^2)$$

THE SECOND SLOT IS THE VARIANCE, not the SD

"mean 70, variance 25"  $\rightarrow \mu=70, \sigma^2=25, \sigma=5$

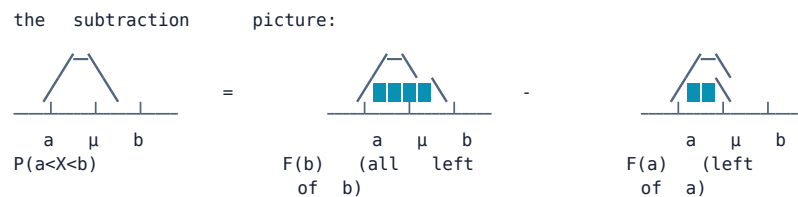
ALWAYS square-root it before a z-score.

$$f(x) = 1/(\sigma\sqrt{2\pi}) \cdot e^{-\{(x-\mu)^2/(2\sigma^2)\}}$$

## 6. The cdf and interval (p012-p013)

$$F(x_0) = P(X \leq x_0) \text{ (area to the left)}$$

$$P(a < X < b) = F(b) - F(a)$$

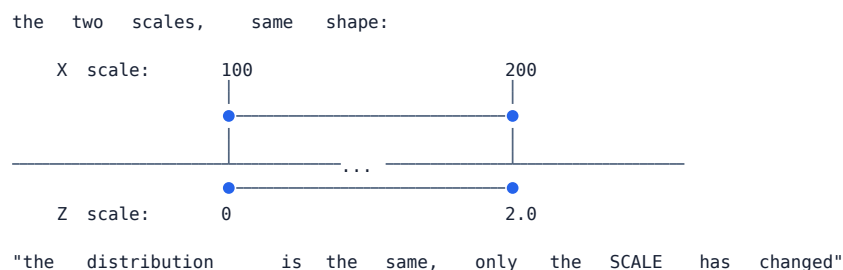


## 7. Standardisation (p014-p015)

$$Z = (X - \mu)/\sigma \quad Z \sim N(0, 1)$$

WORKED:  $\mu=100, \sigma=50, X=200$

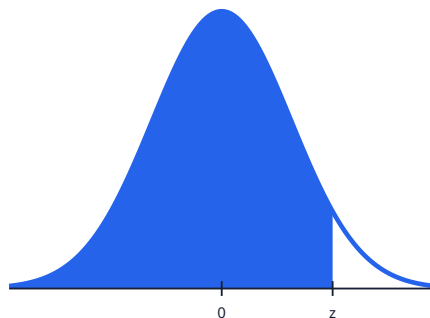
$$Z = (200-100)/50 = 2.0$$



## 8. THE TWO TABLES - the biggest trap in the whole paper (p022-p023)

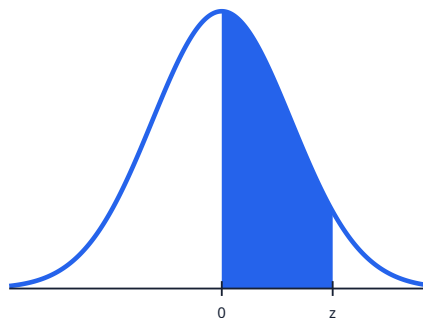
## Two normal-table conventions

p022: cumulative  $P(Z < z)$



$z=2$ : 0.9772

p023: centre area  $P(0 < Z < z)$



$z=2$ : 0.4772

Cumulative and centre-area normal tables

THE SLIDES SHOW TWO DIFFERENT TABLES AND THEY LOOK ALIKE

TABLE 1 (p022, cumulative standard normal table):

gives  $P(Z < z) = \text{CUMULATIVE from } -\infty \text{ to } z$

$F(z)$ . e.g.  $z=0.0 \rightarrow 0.5000$ ,  $z=2.0 \rightarrow 0.9772$

TABLE 2 (p023, area from 0 to  $z$ ):

gives  $P(0 < Z < z) = \text{area FROM THE CENTRE to } z$

$A(z)$ . e.g.  $z=2.0 \rightarrow 0.4772$

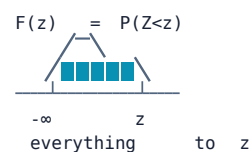
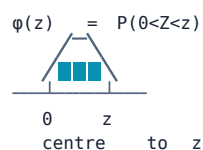
THEY DIFFER BY 0.5 FOR  $z > 0$ :

$F(z) = 0.5 + A(z)$

$0.9772 = 0.5 + 0.4772 \checkmark$

DECIDE WHICH TABLE THE QUESTION GIVES BEFORE SUBSTITUTING

the two areas, drawn:



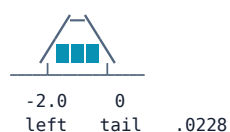
## 9. Symmetry facts (p024-p025)

$P(Z < -2.00) = 1 - P(Z < 2.00) = 1 - 0.9772 = 0.0228$

$\Phi(-z) = 1 - \Phi(z)$

$P(Z > z) = 1 - \Phi(z)$

the mirror picture:



the left tail at  $-z$  EQUALS the right tail at  $+z$

## 10. WORKED: the 8.6 chain (p028-p033)

$X \sim N(8.0, 5.0^2)$  [note:  $\sigma = 5.0$ ]

(a) FIND  $P(X < 8.6)$ :

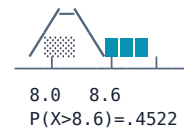
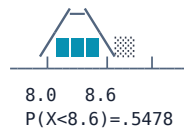
$$Z = (8.6 - 8.0)/5.0 = 0.12$$

$$P(X < 8.6) = P(Z < 0.12) = F(0.12) = 0.5478$$

(b) FIND  $P(X > 8.6)$ :

$$P(X > 8.6) = 1 - 0.5478 = 0.4522$$

the two areas:



ERRATA 8: the figure on p030 labels the curve " $\sigma = 10$ ", but the arithmetic on the same page divides by 5.0. Use  $\sigma = 5.0$ .

## 11. WORKED: the inverse, 20% lower tail (p034-p037)

Find  $x$  so that only 20% of values are BELOW  $x$ .

STEP 1: find  $z$  for a 20% lower tail

look for  $F(z) = 0.20$

from the table,  $z \approx -0.84$  (exactly  $-0.8416$ )

STEP 2: CONVERT BACK TO  $X \leftarrow$  the forgotten step

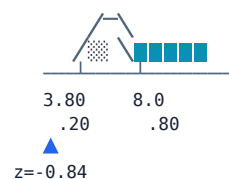
$$X = \mu + Z\sigma = 8.0 + (-0.84)(5.0) = 8.0 - 4.2 = 3.80$$

THE INVERSE HAS TWO STEPS AND PEOPLE STOP AFTER STEP 1.

Getting  $z = -0.84$  is HALF the answer.

The mark is in  $X = \mu + z\sigma$ .

the picture:

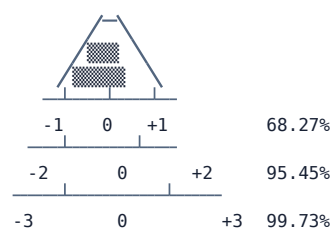


## 12. The landmarks (the hidden item H4)

within  $1\sigma$  : 68.27%

within  $2\sigma$  : 95.45%

within  $3\sigma$  : 99.73%



## PART C: EXPONENTIAL (p038-p043) - lecture 16

### 13. What it models

the LENGTH OF TIME between two occurrences of an event  
(the time between arrivals)  
examples: trucks at a dock, transactions at an ATM, phone calls

### 14. The density and its parameters (p040-p041)

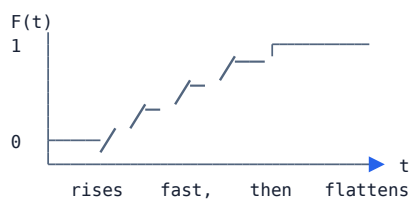
|                                  |                          |
|----------------------------------|--------------------------|
| $f(t) = \lambda e^{-\lambda t},$ | $t > 0$                  |
| $\mu = 1/\lambda$                | $\sigma^2 = 1/\lambda^2$ |
| $F(t) = 0$                       | for $t \leq 0$           |
| $= 1 - e^{-\lambda t}$           | for $t > 0$              |
| $P(T > t) = 1$                   | for $t < 0$              |
| $= e^{-\lambda t}$               | for $t \geq 0$           |

ERRATA 4: the slide on p040 says "1/λ is the MEAN NUMBER of occurrences per unit time". THAT IS BACKWARDS.  
λ = the rate (occurrences per unit time)  
1/λ = the mean TIME between occurrences  
p041 states it correctly as  $\mu = 1/\lambda$ .

THE DECAY PICTURE:



the cdf rises toward 1:



### 15. WORKED: arrivals at 15 per hour (p042-p043)

"customers arrive at the rate of 15 per hour.  
P(the arrival time between consecutive customers < 3 minutes)?"

λ = 15 per HOUR  
3 minutes = 0.05 hours <- CONVERT THE UNIT  
 $P(T < 0.05) = 1 - e^{-\lambda t}$   
 $= 1 - e^{-(15)(0.05)}$   
 $= 1 - e^{-0.75}$   
 $= 1 - 0.4724 = 0.5276$

THE UNIT CONVERSION IS THE WHOLE QUESTION.

$\lambda$  is per hour, so  $t$  MUST be in hours.

3 min = 0.05 hr.

This is the trap the setter builds into almost every exponential story (errata families E24S4, A2 D2).

the mean gap =  $1/\lambda = 1/15$  hour = 4 minutes

## 16. The memoryless property (hidden item H3)

$$P(T > s + t \mid T > s) = P(T > t)$$

"surviving  $s$  MORE, given you already survived to now,  
is the same as surviving  $s$  from scratch"  
the exponential NEVER AGES. A used part is as good as new.  
(unlike, say, a lightbulb that wears out)

PROOF SKETCH:

$$P(T > s + t \mid T > s) = P(T > s + t \text{ and } T > s) / P(T > s)$$

$$= P(T > s + t) / P(T > s)$$

$$= e^{-\lambda(s+t)} / e^{-\lambda s}$$

$$= e^{-\lambda t} = P(T > t) \quad \checkmark$$

the  $e^{-\lambda s}$  cancels <- the whole trick

the canonical number:  $e^{-1} = 0.3679$

## THE THREE CONTINUOUS DISTRIBUTIONS, ONE TABLE

|           | UNIFORM      | NORMAL                            | EXPONENTIAL               |
|-----------|--------------|-----------------------------------|---------------------------|
| pdf       | $1/(b-a)$    | $1/(\sigma\sqrt{2\pi})e^{-\dots}$ | $\lambda e^{-\lambda t}$  |
| mean      | $(a+b)/2$    | $\mu$                             | $1/\lambda$               |
| variance  | $(b-a)^2/12$ | $\sigma^2$                        | $1/\lambda^2$             |
| cdf       | 0, ratio, 1  | table                             | 0 or $1 - e^{-\lambda t}$ |
| tail      | ratio        | $1 - \Phi(z)$                     | $e^{-\lambda t}$          |
| parameter | $a, b$       | $\mu, \sigma^2$                   | $\lambda$ (a RATE)        |
| models    | random pt    | measurement                       | waiting time              |

the three shapes side by side:

UNIFORM



flat  
"equally  
likely"

NORMAL



bell  
"measurement  
error"

EXPONENTIAL



decay  
"waiting  
time"